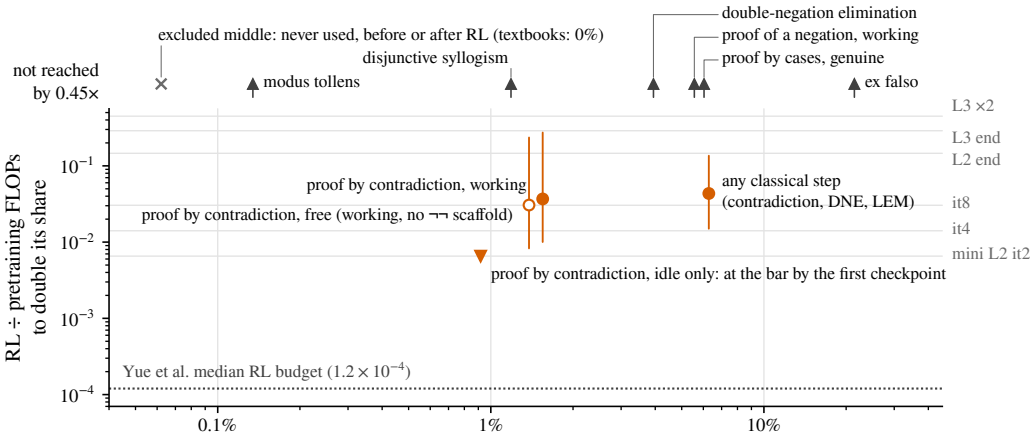
RL needed to double a strategy's share of correct proofs, by how much of it pretraining contained



Natural prevalence, not a knockout. Seed 0, Charles149 (149 problems), k = 1,024. Bar: the strategy's share of all Lean-accepted attempts reaches 2× the base's (2× is our choice). y: R at the first checkpoint of the main run (grey lines) at the bar, interpolated in log R; bars: 95% problem bootstrap; down-triangle: at the bar by the first checkpoint. The low-RL arms (kp16, kdense) are not used. Hollow: free ⊂ working, shifted sideways (1.45 vs 1.47%). The GRPO pools are 40–45% classical-only theorems.